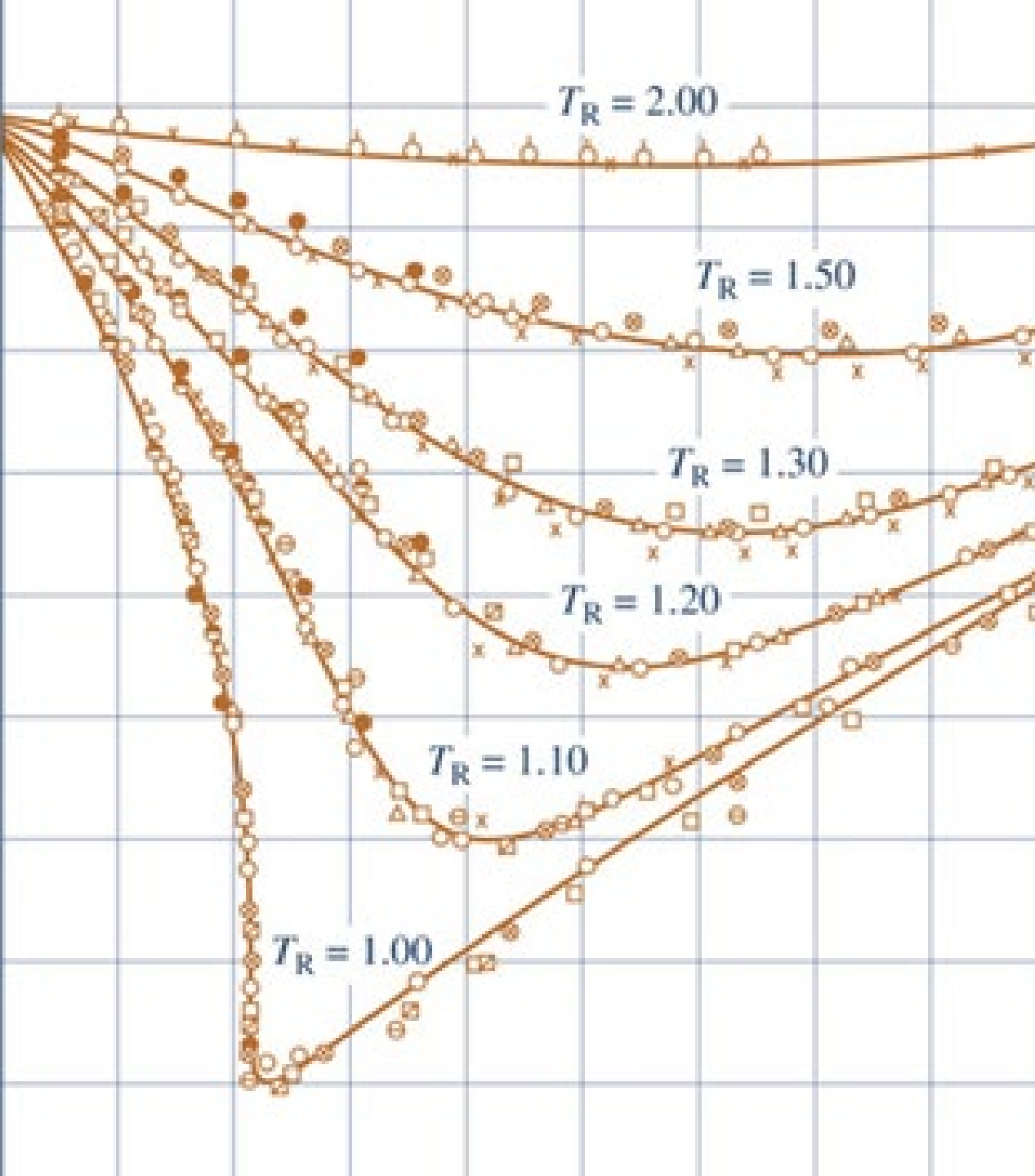


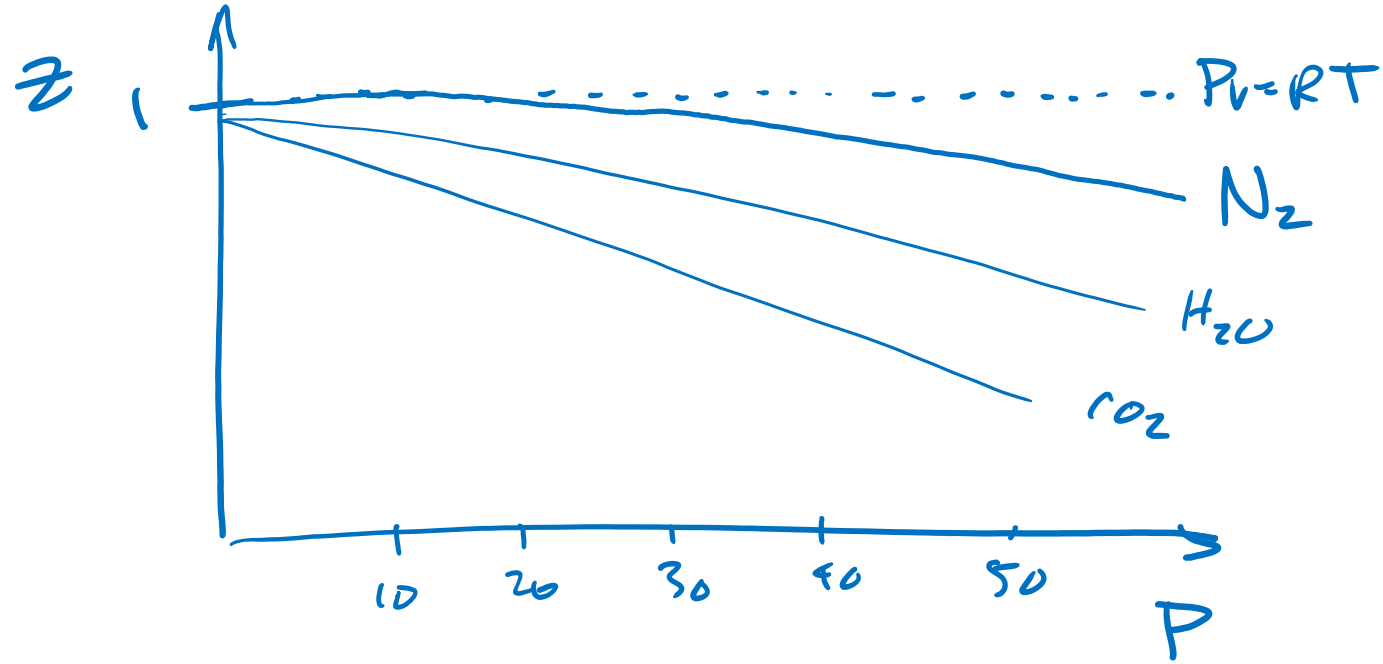
ME 300:
Engineering Thermodynamics

Generalized Compressibility



Generalized compressibility

$$P_v = RT$$
$$z = \frac{P_v}{RT}$$



Example: Generalized compressibility

$$H_2 @ 500K \quad 1 \text{ atm} = 101325 \text{ Pa}$$

$$\rightarrow v = 10.263 \text{ m}^3/\text{kg}$$

$$\rightarrow R = \frac{R_u}{M} = 2078.62 \frac{\text{J}}{\text{kg} \cdot \text{K}}$$

$$Z = \frac{Pv}{RT} = \frac{(101325)(10.263)}{(2078.62)(500)} \\ = 1$$

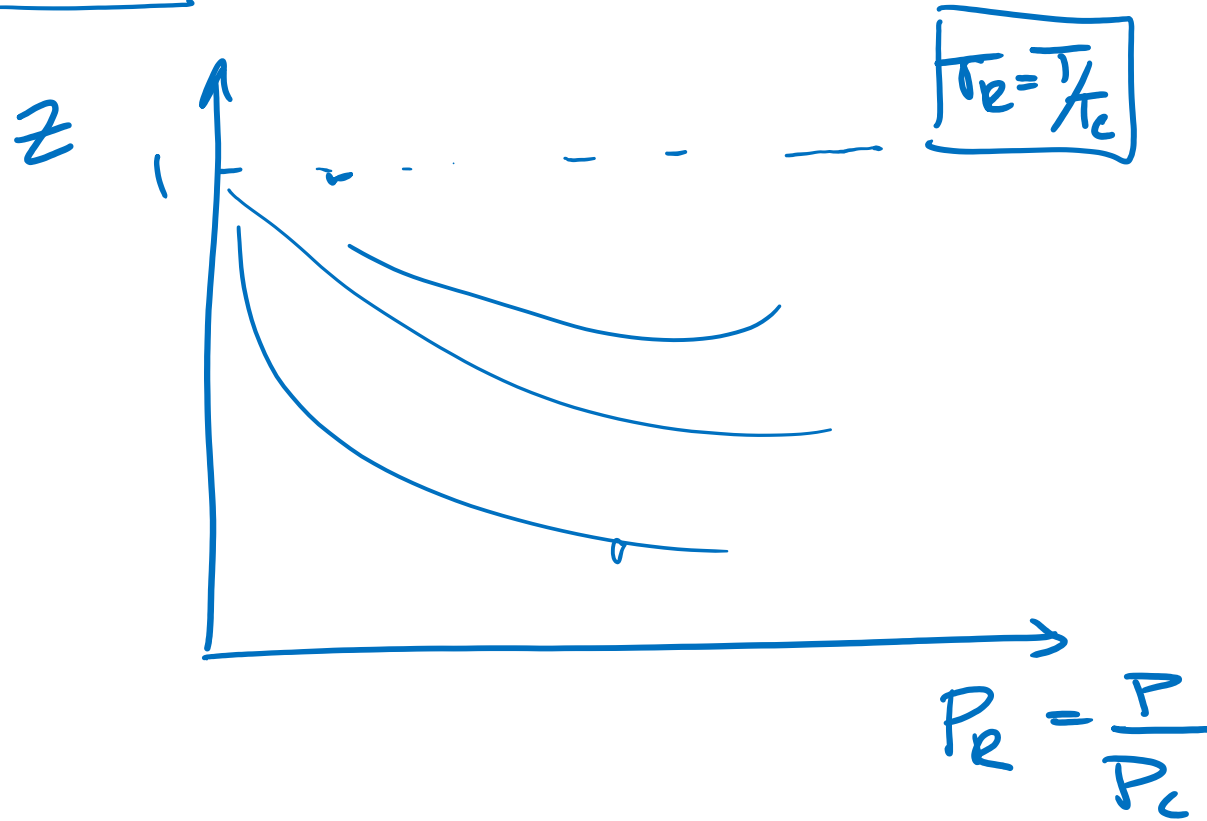
$$H_2O \quad 500K \quad 2 \text{ MPa}$$

$$\underline{Z} = 0.9$$

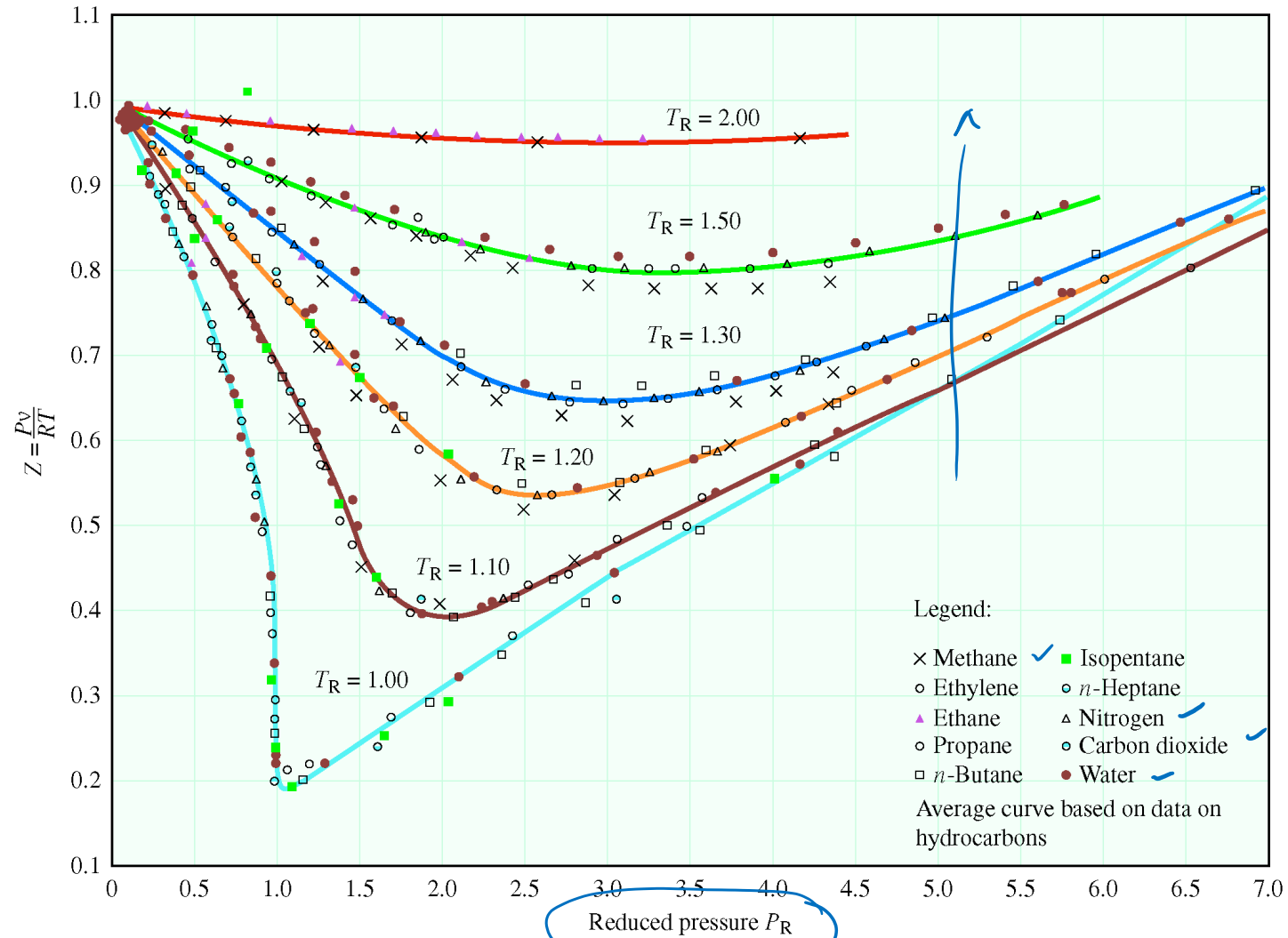
↓

Critical point

P_c T_c $\rightarrow$ liquids and gases are no longer distinguishable



Compressibility chart



The compressibility factor Z can be correlated using reduced properties (i.e., P/P_c and T/T_c). Note how the data for various substances collapse when plotted in this manner. Adapted from Ref. [12] with permission.